# DANG TRAN NAM

Software Engineer · Ho Chi Minh City, Vietnam

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Software Engineer specializing in backend systems, high-concurrency database architectures, quantitative trading engines, and production full-stack SaaS. Experienced in designing resilient distributed edge agents, low-latency transaction engines, algorithmic backtesting frameworks, and enterprise .NET web systems.

## Experience

### An Binh Hospital

Software Engineer Intern — IT Asset Management (ITAM): Distributed Telemetry System

- **Distributed Telemetry Agent:** Designed and developed a cross-platform Go agent harvesting real-time CPU, RAM, Disk, and top processes via `gopsutil v3` across 100+ edge devices for centralized observability.
- **Resilience & Reliability:** Implemented Circuit Breaker pattern (100 req/s limit on server) with configurable jitter (0-30s) and exponential backoff (5s → 10s → 20s) to absorb thundering-herd reconnect storms, maintaining 99.9% uptime.
- **Performance Optimization:** Reduced agent memory footprint by over 60% (under 15MB); built automated log rotation (10MB x 7), startup SQLite database backups, and 30-day historical data pruning.
- **Security & Management:** Secured endpoint telemetry with HTTP Bearer token authentication and remote kill switch (sleep/exit commands); deployed via Windows GPO scheduled tasks and load-tested with 500 simulated agents.

## Key Projects

## Radixify — Full-Stack AI Linguistic SaaS

**Problem:** Memorizing vocabulary without understanding etymological structure leads to rapid attrition.

**Solution:** Built full-stack SaaS with 3-pillar breakdown: Morpheme DNA decomposition, SVG Bézier etymology trees, and physics-based D3.js synaptic neuron maps. Implemented dynamic AI waterfall (Gemini 2.5 Flash → DeepSeek V3 → GPT-4o-mini → Groq) with morphotactic state machine fallback, sub-5ms 2-tier cache (L1 LRU + L2 PostgreSQL), SM-2 Spaced Repetition flashcards, automated PayOS VietQR + Polar.sh/PayPal checkout, and Remotion video generation pipeline.

**Result:** Live in production at radixify.app with 155 passing Vitest tests and sub-second latency.

Live Demo: radixify.app · GitHub Repository

## High-Concurrency Inventory Engine

**Problem:** Peak flash-sale traffic causes extreme concurrency race conditions and inventory overselling.

**Solution:** Engineered Single-Writer / Multi-Reader (SWMR) model with 100-connection reader pool, 1-connection writer queue guarded by Tx Mutex, and strict serializability (Stock = 1). Integrated PayOS QR payment webhook handling, LocalStack AWS S3, and K6 load testing suite.

**Result:** Benchmarked at ~10,000 writes/sec on a single SQLite node with p95 latency < 0.1ms and zero overselling. Graduation thesis project.

GitHub Repository

## Zone Swing Trading Engine

**Problem:** Discretionary crypto swing traders suffer from emotional execution, delayed signal detection, and unverified indicator logic.

**Solution:** High-performance algorithmic crypto trading engine (~46,000 LOC Go, 31 CLI commands, 27 internal packages, 78 tests) reverse-engineering the PivotZone model with strict parity validation against TradingView signals. Implemented multi-tier pivot prominence active zone selection, FVG/regime filters, automated Telegram alert daemon, state persistence, and comprehensive multi-asset backtest cache (BTC, ETH, SOL, SUI, XRP).

**Result:** Deployed as 24/7 background daemon on Fly.io/VPS with automated state persistence and real-time execution.

GitHub Repository

## GoBacktest — Quantitative Strategy Evaluation Framework

**Problem:** Crypto backtests typically overfit historical data in notebook scripts with no execution stress, no slippage modeling, and zero out-of-sample discipline.

**Solution:** Open-source Go backtesting and stress-testing framework featuring chronological multi-symbol simulation (fill at next open, leg slippage), Walk-Forward Analysis, Leave-One-Symbol-Out cross-validation, and stress testing (slippage, funding APR decay, +1 candle delay). Built deterministic seedable Monte Carlo trade shuffling (5,000+ iterations) for empirical drawdown distributions with zero external dependencies.

**Result:** Reproducible, institutional-grade strategy evaluation delivering precise win rate, profit factor, CAGR, and drawdown metrics.

GitHub Repository

## Quant Trading Monorepo — Vault 8386

**Problem:** Capture non-directional statistical arbitrage across traditional securities and crypto derivatives.

**Solution:** Dual-engine automated trading system: (1) Cash & Carry delta-neutral funding rate arb on Binance Futures (Spot Long + Short Perp x1, locking funding APR > 119% with kill-switch on yield decay < 10%); (2) HOSE ETF I-NAV arbitrage & automated DCA for VNDIAMOND ETF (FUEVFVND) via DNSE API on GCP Cloud Run.

**Result:** 24/7 cloud execution with zero directional price risk.

GitHub Repository

## TapHoaNho — Retail Store Management ERP

**Problem:** Retail multi-branch operations require unified batch inventory tracking, supplier management, dynamic promotion policies, and POS checkout.

**Solution:** ASP.NET Core 9 Web API using Clean Architecture, Repository & Unit of Work pattern, AutoMapper, FluentValidation, and JWT auth. Frontend built with React 19, TypeScript, TanStack Router (type-safe routing), TanStack Query, Zustand, Ant Design, camera QR code product scanning, and draft order handling.

**Result:** Strict domain isolation, end-to-end transactional consistency across orders/stock, and zero frontend runtime route mismatches.

GitHub Repository

## minPOS — Offline-First Micro-Merchant POS

**Problem:** Small retail/F&B merchants need an ultra-fast, affordable POS system that works reliably without internet.

**Solution:** Embedded Go backend (Gin, modernc.org/sqlite WAL mode, JWT auth, hardware-bound license activation CLI) paired with Flutter mobile/tablet client. Features responsive product grid, CartController state management, bill preview modal, order history, daily sales reporting, and Excel data export.

**Result:** Sub-50ms query latency, offline-first reliability, zero sales data loss.

Local MVP · Private Repo

## Technical Skills

**Languages:** Go (Golang), TypeScript, JavaScript, C# (.NET 9), Python, Dart, SQL, C/C++, Rust

**Frameworks & Backend:** Go Fiber v3, Gin, ASP.NET Core 9, Next.js 15/16, React 19, Node.js, Express, Vite, Flutter, Prisma ORM, Entity Framework Core, HTMX, Tailwind CSS v4, TanStack Router & Query

**Databases & Caching:** PostgreSQL, SQLite (WAL Mode), Redis, Neon Serverless

**Architecture & Distributed Systems:** High-Concurrency Architecture (SWMR), Circuit Breakers, Quantitative Backtesting Engines, Walk-Forward Analysis, Monte Carlo Simulation, Clean Architecture, Repository & Unit of Work, SM-2 Spaced Repetition, REST APIs, WebSockets

**DevOps & Infrastructure:** Linux, Docker, GCP (Cloud Run, Compute Engine), Fly.io, Vercel, Git, K6 Load Testing, Vitest, LocalStack, Puppeteer

## Education

**Saigon University (SGU)** — Software Engineering (2021 - 2026)
Engineer Degree

## Certifications

- TOEIC (760 R/L)
- AWS Cloud (In Progress)